

	Statement D is not an essential assumption of kinetic theory of gas and can be derived from pressure of a gas
10.	Ans: C heat gained by ice in melting + heat gained by 20 g of water (total mass of water) = heat lost by 130 g of water $0.020 \times 3.3 \times 10^5 + [0.020 \times 4.2 \times 10^3 \times (T - 0)] = 0.130 \times 4.2 \times 10^3 \times (50 - T)$ $6.6 \times 10^3 + 84 T = 27.3 \times 10^3 - 546 T$ $630 T = 20700$ $T = 33 \text{ } ^\circ\text{C}$ distractor: $T = 38 \text{ } ^\circ\text{C}$ (heat gained by melted ice water omitted)
11.	Ans: C